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MILITARY SPECIFICATION

PRINTER, CONTACT, PHOTOGRAPHIC CONTINUOUS EN-6B

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force

*1. SCOPE

1.1 This specification covers one type of automatic, continuous contact photographic printer for printing either black and white or color, designated EN-6B.

*2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-636	Motor , Alternating-Current, (Fractional Horsepower)
NN-P-530	Plywood, Flat Panel
QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet
QQ-A-601	Aluminum Alloy Sand Castings
TT-E-529	Enamel, Alkyd, Semi-Gloss
TT-V-109	Varnish, Spar, Alkyd-Resin
PPP-B-35	Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated

Military

MIL-F-32	Film, Photographic, Aerial, Black and White
MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material, Water Vaporproof Flexible, Heat Sealable
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-3767	Connectors, Plug and Receptacle, Electrical (Power, Bladed Type), General Specification for
MIL-P-4672	Paper, Photographic, Black and White, Sensitized
MIL-D-4826	Developer, Photographic, Low Contrast, Armed Forces Developer No.3

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MIL-D-4827	Developer, Photographic, Paper, Armed Forces Developer No 25
MIL-S-5059	Steel, Corrosion-Resistant (18-8), Plate, Sheet and Strip
MIL-C-5756	Cable and Wire, Power, Electric, Portable
MIL-S-6721	Steel, Corrosion-and Heat-Resistant (Chemically Stabilized) Plate, Sheet, and Strip
MIL-M-7793	Meter, Time Totalizing
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base

STANDARDS

Federal

FED-STD-595	Color

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-150	Photographic Lenses
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment
MIL-STD-200	Electron Tubes, Selection and Use of
MIL-STD-415	Test Points and Test Facilities for Electronic Systems and Associated Equipment, Design Standard for
MIL-STD-461	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-681	Identification Coding and Application of Hookup and Lead Wire
MIL-STD-808	Finishes, Protective, and Codes, For Finishing Schemes For Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1278	Filters, Light, Photographic
MS26565	Spool, Photographic, Film, Aerial
MS33586	Metals, Definition of Dissimilar

DRAWINGS

U.S. Air Force

55K17647	Printer - Contact, Photographic, Continuous, Type EN-6 Outline of

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Electrical Manufacturer's Association Standard

MG-1 Motors and Generators

(Application for copies should be addressed to the National Manufacturer's Association, 155 East 44th Street, New York, New York 10017.)

United States of America Standards Institute

USAY32.16-1968 Electrical and Electronic Reference Designations

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

*3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing (see 4.4).

3.2 Components. The printer shall consist of the following major components:

ITEM NO	QUANTITY	DESCRIPTION	SEE REQUIREMENT
1	1 each	Main Frame and Housing	3.8.1
2	1 each	Printing Light System	3.8.2
3	1 each	Film Transport System	3.8.3
4	1 each	Automatic Exposure Control	3.8.4
5	1 each	Printer Controls	3.8.5

*3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.4 Materials

3.4.1 Metals. Metals shall be of the corrosion resistant type or treated to resist corrosion due to photographic chemicals, or atmospheric conditions likely to be met in storage or normal service.

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3.4.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.1.2 Corrosion-resistant steel. Unless otherwise specified, all untreated steel parts, including accessories, shall be fabricated of corrosion resistant steel as follows:

*3.4.1.2.1 Welded parts. All parts fabricated by welding, to any extent, shall be made of steel conforming to MIL-S-6721.

*3.4.1.2.2 Untreated parts. Sheathing, trim, outer wall covering, and unwelded parts, shall be made of steel conforming to composistion 302 of MIL-S-5059.

3.4.1.3 Aluminum alloy castings. When used, such castings shall comply with QQ-A-601.

*3.4.1.3.1 Aluminum parts. Parts fabricated of aluminum shall have their composition in accordance with QQ-A-250/5 with maximum corrosion resistance.

*3.4.2 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a suitable fungistatic agent. Overall spraying or brushing of the printer with fungicide shall not be permitted. Fungistatic coating material, if used, shall be Type I varnish in accordance with MIL-V-173. Fungus-resistance tests are not required, unless requested by the procuring activity, and when tests are not conducted, the contractor shall furnish certification, listing non-inert materials and their physical location in the printer.

*3.4.3 Protective treatment. When materials are used that are subject to deterioration when exposed to environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Any protective coating that will crack, chip or scale when subjected to the tests of this specification shall not be used.

*3.5 Design. The printer shall be designed as a self-contained unit for automatic continuous printing of 70 millimeter (mm), 5- and 9 1/2-inch aerial camera negatives in rolls up to 500 feet long. It shall satisfactorily expose 9 3/8-inch wide rolls of projection, graded, contrast paper conforming to MIL-P-4672, Type II, subtype E; 9 3/8-inch wide rolls of contact, variable, contrast paper conforming to MIL-P-4672, Type IV, subtype E; and 70mm, 5- and 9½-inch rolls of aerial duplicating film conforming to MIL-F-32, designated 14-14-24d, in lengths of up to 500 feet. The printer shall accomodate film spools conforming to MS26565 and having nominal capacities of up to 500 feet.

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The printer shall be capable of printing and tracking the above plus thin base roll materials conforming to MIL-F-32, at any of the three speeds of operation specified in 3.5.2. In addition, the design shall provide for a 5-digit running time meter, conforming with MIL-M-7793, to indicate the cumulative actual operation hours, to the nearest 0.1 hour, up to 9999.9. The actual maximum capactiy of the printer, not to exceed 500 feet, shall be as specified by the procuring activity. (see 3.6 and 6.2)

3.5.1 Safety. Safe design shall include the prevention of reversed assembly or installation of connectors, fasteners, parts and units where possible malfunction and personnel hazards might occur.

*3.5.2 Speed of operation. The printer shall be capable of continuously printing 70mm, 5- and 9½-inch roll materials at any of 3 fixed speeds of 5, 20, or 60 feet per minute (fpm).

*3.5.3 Mode of operation. The printer, when loaded with negative and sensitized printing materials, shall continuously drive and spool the 2 materials at the 3 fixed printing speeds. The materials shall be brought into close contact at the printing aperture. The proper exposure, which has an average density of not greater than 2.1, shall be produced by means of an automatic exposure control system that continually scans the negative material previous to printing. The printer shall contain the necessary override, dodging, and calibrating controls used in printing abnormal types of negatives, and for calibrating the printer to the sensitivity of the printing material being used.

*3.5.3.1 Filters. The printer shall be equipped with a filter system to allow printing on variable contrast materials. The filter system will consist of either a set of insertable geletin type filters (see 3.8.2.4) or a roll type filter (see 3.8.2.5), as specified (see 6.2).

3.5.4 Film and paper damage. The printer shall not mar, scratch, tear, or otherwise damage the original or duplicating material.

3.5.5 Exposure control. The printer exposure control system, consisting of a negative reading system, meter, and automatic exposure control shall be capable of measuring and printing neutral densities from 0.3 to 2.1 (representing average negative densities of 0.3 to 2.1), and of controlling the level of exposure so that any of these densities print to essentially the same print density of ± 0.10. This performance shall be obtained when printing 9 3/8-inch roll paper in accordance with that specified in 3.5. The above tolerance is applicable for any 1 printing paper after the printer has been calibrated to obtain a basic exposure while operating on midrange of the automatic exposure control at 60 feet per minute, and using a 1.0 ± 0.05 neutral density as a negative to obtain a reflected print density of 0.70 ± 0.05.

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3.5.6 Uniformity of illumination. The illumination across the printing aperture of the printer shall be uniform within a range of ± 5 percent.

3.5.7 Contact. The printer shall maintain sufficient and uniform contact between the negative and printing material over the entire width of the printing slit and for any length of material exposed up to 500 feet.

3.5.8 Resolution. The printer shall be capable of reproducing high contrast resolution targets of at least 90 lines per millimeter any where across the printing aperture.

3.5.9 Maintenance. The printer shall have a minimum number of parts consistant with reliability and shall be contained in a single assembly or minimum number of assemblies. The design shall utilize readily removable and replaceable units to facilitate easy assembly, disassembly, location of trouble sources, and repair with a minimum "down time" of the printer. The design shall be such that adjustments, minor repairs, and replacement of parts may be made by service personnel with a minimum of training, using common commercial tools. Any part of a drive mechanism requiring timing shall be provided with timing marks.

3.5.10 Test points and built-in test facilities. Test points and built-in test facilities shall be included in the design of the equipment. Electronic test points shall be in accordance with MIL-STD-415. Mechanical test points, if needed, shall be provided. They shall be simple in design, shall not cause protrusions on the equipment, and shall require a minimum of accessory equipment or special tools.

*3.6 Construction. All parts of the equipment shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and operation without deterioration, malfunction, or any other damage that will affect its use. At the option of the procuring activity, a procurement model of the equipment may be made available for inspection by prospective bidders and loaned to the contractor. In this case, physical construction shall conform to the model, however the performance characteristics shall be as specified herein and not necessarily as represented by the model. (see 6.2)

*3.6.1 Lubrication. The printer shall be lubricated prior to testing and shipment. Where it is necessary to use lubricants such as silicone greases, molybdenum disulfide, or colloidal suspensions of graphite, approval shall be obtained from the procuring activity.

3.7 Performance.

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*3.7.1 Environmental conditions. There shall be no failure or impairment of function when the printer is operated under the following conditions:

a. Temperature

(1) Ambient temperatures from 40 degrees (°) Fahrenheit (F) to 115°F.

(2) Subsequent to extended storage (6 months or longer) at any or all temperatures within the range from -80°F to +160°F.

b. Humidity. Relative humidities of up to 97 percent for periods of up to 20 hours; or after exposure at 100 percent humidity, with condensation, for periods of up to 5 hours.

c. Elevation.

(1) Operation at altitudes up to 10,000 feet above sea level.

(2) Subsequent to shipment or storage at altitudes up to 50,000 feet above sea level.

d. Vibration. Vibration within the range of 2 to 55 cycles per second.

*3.7.2 Interference. Interference control requirements shall conform to MIL-STD-461. The equipment class shall be II b.

3.7.3 Dielectric. All circuits of the printer shall be capable of withstanding the conditions herein. There shall be no flashover or breakdown when the rated test voltage, specified herein, is applied between any electrical circuit and ground. All adjacent metal parts not in the circuit under test shall be considered as ground.

3.7.4 Leakage current. Following the exposure to the humidity and temperature conditions specified herein, the leakage current between all electrical circuits and ground shall not exceed a total of 2 milliamperes. All adjacent metal parts not in the circuit under test shall be considered as ground.

3.8 Details of components.

3.8.1 Main frame and housing. The printer main frame and housing shall be fabricated in such a manner as to provide strength and rigidity to maintain all working parts and assemblies in true alignment. The printer body shall be fabricated primarily of aluminum plate and alloy castings supplemented with fabricated sheet metal assemblies.

3.8.1.1 Compartment panels. Where it is desired to have access to the interior, compartment panels shall be constructed so that they are readily removable and accessible for ease of maintenance of the printer.

*3.8.1.2 Viewing safelight. Provisions shall be made for viewing the negative material as it passes through the printer. The viewer shall be approximately 5 x 10 inches in size and located so that when the negative is viewed, the dodging controls, contrast control, or other operating controls can be activated immediately thereafter if the operator so desires. The viewing safelight shall contain safelight filters to prevent fogging of sensitized materials during printing when using materials as specified herein.

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3.8.1.3 Instrument panel lights. Provisions shall be made for lights to illuminate the instrument panel markings, operating controls and exposure meter. These instrument panel lights shall be adequately protected to prevent fogging of the sensitized materials during processing.

*3.8.2 Printing light system. The printing light system shall provide uniform illumination across the entire printing aperture.

3.8.2.1 Printing lamp. The printing lamp shall be mounted in a sturdy metal lamphouse. The lamphouse shall be blower cooled to eliminate any possibility of excessive heating. The printing lamp shall be a 115 volt medium prefocused lamp of sufficient wattage to satisfy the test requirement specified herein. The printing lamp shall be readily accessible for replacement.

3.8.2.1.1 Printing lamp adjustment. The base shall be provided with three adjustments so that the lamp filaments may be properly centered on the optical axis of the printer. There shall be a horizontal adjustment, a vertical adjustment and an adjustment along the optical axis of the printer.

3.8.2.2 Collimating lens system. A collimating lens system shall be provided so that a light beam, consisting of essentially parallel rays and of uniform intensity, is provided over the entire printing aperture. The lens system shall permit installation of a servo-controlled exposure control in the light path to provide an accurate means of controlling light intensity at the printing aperture as demanded by the light metering circuit and phototube.

3.8.2.3 Blower. A blower shall be provided to maintain the printing light system at a safe operating temperature. The temperature of the filter pack at the filter carrier shall not exceed 30°F above an ambient range of 40°F to 120°F while the printer is in continuous operation.

*3.8.2.4 Insertable filters. When specified as insertable type, the filter system shall consist of filter carriers and contrast filters as follows:

*3.8.2.4.1 Filter carriers. Provisions shall be made for placing gelatin color correction filters and neutral density filters in the light path, as close to the light source as possible, but not closer than the heat absorbing glass. The filter holder shall be equipped with a retaining window and two sheets of clear flat glass that will hold one or more gelatin filters securely over the aperture in the filter carrier. The filter carrier shall be inserted into the printer on a track which positions the filter carrier correctly in the light path. The filter carrier shall be located in the light path so that any foreign particles such as dust, scratches, fingerprints, etc., that appear on the filters shall not be in sharp focus at the printer aperture, causing uneven illumination.

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*3.8.2.4.2 Contrast filters. Each printer shall be supplied with a series of five (5) variable contrast printing filters conforming to MIL-STD-1278, Type I, Grade C, Class P. The filters supplied shall be Eastman Kodak Co Polycontrast or approved equal, grades 1, 1 1/2, 2, 3, and 4, and conforming to the Military Filter Designations of P-2-1P, P-2-1.5P, P-4-2P, P-4-3P, and P-4-4P, respectively. Each filter shall be separately mounted in a holder which will protect the filter from any damage such as dust, scratches, fingerprints, etc, which might occur during normal usage. The contrast filters shall be correctly positioned in the light path by means of a locating track or guide. A storage compartment for the filters shall be provided in a convenient location on the printer.

*3.8.2.5 Roll type filters. When specified as roll type, the filter system shall consist of a filter roll, guide and take-up rollers, and a filter transport mechanism as follows:

*3.8.2.5.1 Filter roll. The filter shall be a roll of DuPont Mylar or Cronar, or equivalent, plastic material with five (5) filter areas, Dupont Varigram or approved equal, grades 1, 3, 5, 7, and 10 conforming to Military Filter Designations of P-2-1V, P-2-3V, P-5-5V, P-4-7V, and P-4-10V, respectively, dyed into it, plus a clear area on each end. Each filter area shall be 3 inches by 5 1/2 inches in size. No white light shall leak around the edges. The roll filter shall be carried in a moisture-proof bag during initial shipment and storage.

*3.8.2.5.2 Guide rollers. There shall be a series of rollers to properly guide the roll filter from the supply roller; in the printing aperture; to the take-up roller. These rollers shall protect the filter from tearing and scratching. The filter shall be placed in a position where the heat from the printing lamp will not harm it after long usage and yet at a place where the light is sufficiently diffused so that any minor defects or dirt on the filter will not adversely affect the print.

*3.8.2.5.3 Take-up roller. The roll filter shall be attached to the take-up roller in such a manner that it will not be inadvertently pulled loose during use, but so that it may be installed and removed when placing the printer into and out of use.

*3.8.2.5.4 Filter transport mechanism. The filter roll shall be transported into the printing aperture by turning hand cranks. A filter position indicator shall be provided to index the filter area in the printing aperture.

3.8.2.6 Heat absorbing glass. The printing lamphouse shall contain a heat absorbing glass located in the light beam which shall retain a larger portion of the heat emitted from the printing lamp within the lamphouse and protect the filter holder from excessive heat. The heat absorbing glass shall be readily accessible for replacement and cleaning.

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3.8.2.7 Color temperature. The color temperature of the light delivered, when connected to 115 ± 10volt power source, shall be between 2850° Kelvin - 3200° Kelvin (K) when measured at the printing aperture with a "Spectra" color temperature meter made by the Photo Research Corporation, or approved equal.

3.8.3 Film transport system. The film transport system shall contain the supply and take-up spindles, idler and tension rollers, along with the printing focal roller, to drive and hold the positive and negative materials in close contact at the printing aperture. All drive tension, and idler rollers, along with the printing focal roller, shall have finishes to prevent any scratching or abrasion of the positive or negative materials.

3.8.3.1 Printing focal roller. The film transport system shall have a printing focal roller that shall be used to bring the negative and positive materials into close contact as they pass over the printing aperture. The drive of the printing focal roller shall be smooth and continuous at the three fixed speeds of operation in order to prevent any slippage, skid, poor tracking, or poor contact between the negative and positive materials. The printing focal roller shall be designed to handle 70mm, 5- and 9 1/2 inch materials.

3.8.3.1.1 Tension rollers. The film transport system shall be provided with tension rollers to produce proper drive and tension of the positive and negative materials in order to maintain optimum contact of the film materials at the printing aperture. All tension rollers shall be carefully aligned and shall present a minimum of friction to the passage of the film. The tension rollers shall be designed to handle 70mm, 5- and 9 1/2 inch materials.

3.8.3.1.2 Idler rollers. The film transport system shall include a sufficient number of idler rollers to confine the positive and negative materials to the proper path. All rollers shall be perfectly aligned and shall be designed to handle 70mm, 5- and 9 1/2 inch materials.

3.8.3.2 Positive feed path. The positive feed path shall be located in the upper carriage assembly or upper portion of the printer body. The film moving mechanism shall transport the positive material from the supply spool over the printing focal roller onto the take-up spool.

*3.8.3.2.1 Positive supply. A positive film supply shall be provided that will accept rolls of 9 3/8-inch printing paper wound on 3 3/4-inch diameter wood cores, and rolls of 70mm, 5- and 9 1/2-inch film duplicating materials wound on spools conforming to MS26565. Sufficient drag or back tension shall be maintained to insure that the positive material is tightly wrapped on the supply spool.

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*3.8.3.2.2 Positive take-up. A continuous type take-up shall be provided to maintain tension on the material regardless of the amount of positive film on the take-up spindle. The positive take-up shall accept rolls of 9 3/8-inch printing paper wound on 3 3/4-inch diameter wood cores, and rolls of 70mm, 5- and 9 1/2-inch film duplicating materials wound on spools conforming to MS26565. Sufficient tension shall be maintained to insure that the positive material is tightly wrapped on the take-up spindle.

*3.8.3.2.3 Positive flange adapters. Two sets of flange adapters shall be supplied for the positive supply and take-up spindles. The flange adapters shall be used to support lengths of printing materials.

*3.8.3.2.4 Positive spool adapters. The positive supply and take-up shall be supplied with a spool adapter for supporting film materials wound on standard 70mm and 5-inch spools conforming to MS26565.

*3.8.3.2.5 The material capacity of the positive section shall be the maximum capability as specified by the procuring activity (see 3.5 and 6.2).

3.8.3.3 Negative feed path. The film moving mechanism shall transport the negative material from the supply spool over the viewing lamp, across the negative reading station, and over the printing aperture onto the take-up spool.

*3.8.3.3.1 Negative supply. A negative film supply shall be provided that will accept rolls of 70mm, 5- and 9 1/2-inch negative materials wound on spools conforming to MS26565. Sufficient drag or back tension shall be maintained to insure that the negative material is tightly wrapped on the feed spool.

3.8.3.3.1.1 Negative rewind. A negative rewind shall be located on the negative supply spindle to be used by the operator to rewind the negative material where more than one printing is required.

*3.8.3.3.2 Negative take-up. A continuous type take-up shall be provided to maintain tension on the material being taken-up regardless of the amount of negative film on the take-up spindle. The negative take-up shall accept rolls of 70mm, 5- and 9 1/2-inch negative materials wound on spools conforming to MS26565. Sufficient tension shall be maintained to insure that the negative material is tightly wrapped on the take-up spindle.

*3.8.3.3.3 Negative spool adapters. The negative supply and take-up shall be supplied with a spool adapter for supporting film material wound on 70mm/5-inch spools in accordance with MS26565.

3.8.3.3.4 Negative footage indicator. A footage indicator shall be provided indicating the total negative footage run through the machine. The indicator shall read up to 999 feet and be resettable to zero after each printing run. The indicator shall be activated by a metering roller located as near to the printing aperture as possible.

*3.8.3.3.5 The material capacity of the negative section shall be the maximum capability as specified by the procuring activity (see 3.5 and 6.2).

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3.8.4 Automatic exposure control system. The printer shall have an automatic exposure control system that will control the level of illumination, at the printing aperture, that is required to properly print any negative which has an average density of not greater than 2.1. This control shall be achieved by the use of a pair of continuous tone neutral density glass wedges located in the printing light beam, varying from low density to high density. One or both of the wedges shall be servo-driven through an amplifier and phototube system. The neutral density wedges shall be readily removable for cleaning and shall be located in the light path so that any foreign particles such as dust, scratches, fingerprints, etc, shall not appear in sharp focus at the printing aperture to produce uneven illumination. The exposure control system shall be usable with 70mm, 5- and 9 1/2-inch negative materials. The control shall operate over the full range of the exposure wedges at any of the 3 printing speeds, specified in 3.5.2, when the printer is being operated in an automatic manner.

3.8.4.1 Negative scanning system. The negative material shall be scanned or read previous to printing in a continuous fashion and the responses received from the phototube shall continually control the position of the neutral density wedges which in turn will control exposure. The exposure control system shall be designed so that the area of scan may be shifted from one edge of the negative to the other edge to allow for selective area scan of the negative material. The negative scanning system shall determine average density by an averaging photocell circuit. The photocell arrangement shall, in effect, be a transmission type densitometer.

3.8.4.1.1 Exposure meter. The negative scanning system shall contain an exposure meter control located to the right side of the viewer assembly. This control shall provide a visual indication of the operation of the automatic exposure control system. The exposure indicator shall show the amount by which the exposure is being changed during operation of the printer. The calibrations on the exposure meter shall be an arithmetical series of numbers (example, 1, 2, 3, 4, 5, etc) indicating the doubles of exposure. The half double positions shall also be marked but not numbered. The exposure meter shall have range to meet the performance requirements as specified herein and also cover the 3 fixed printing speeds of operation.

3.8.5 Printer controls.

3.8.5.1 Dodging controls. Calibrated dodging controls shall be conveniently located at the front of the printer to permit independent variation in light intensity at the sides and center of the printing aperture. The calibration on the dodging control shall be a geometrical series of numbers (example 1, 2, 4, 8, 16, etc) indicating the double and half-double exposure.

3.8.5.2 Exposure override control. To properly print abnormal negatives such as those of wooded areas, snow scenes, or desert area, an exposure override shall be provided to permit over-exposure or under-exposure as desired. The control shall be graduated in double and half-double increments of exposure with a range of ± 3 doubles around the point for printing normal negatives. This range of exposure override shall be part of the range of the automatic exposure control.

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3.8.5.3 Emulsion sensitivity control. The printer shall have an emulsion sensitivity control which shall be used to calibrate the printer to the speed of the printing material being used. This control shall be part of the automatic exposure control system and shall be used to change the position of the neutral density glass wedges, thus changing exposure and calibrating the printer to the speed of the material being used. The control shall be graduated in doubles and half-doubles of exposure with a range of ± 1 1/2 doubles around the point for printing normal speed materials.

3.8.5.4 Drive speed control. The printer shall be provided with a drive speed control to control the rate of speed of the negative and printing materials being fed through the printer. The control shall read in feet per minute and shall have 3 fixed speed settings, 5, 20, and 60 feet per minute.

3.8.5.5 Power switches. The printer control panels shall have power switches for the printing lamp, film drive, panel lights, automatic exposure control, viewer lights, main power, and a manual exposure momentary switch used for manual operation of the exposure control system. The panel lights shall also have a rheostat control for regulating the intensity of these lights.

*3.9 Case. When a case for the printer is specified by the procuring activity, (see 6.2) it shall be made of watertight, reinforced materials. The case, as specified herein, shall be constructed of plywood conforming to NN-P-530, CS 35, type I, grade 3-4. All external edges and corners shall be reinforced with formed sheet-steel angles not less than 0.040 inch thick and secured with flathead corrosion-resistant screws or rivets. The flathead screws shall be flush with the surface. The metal corner reinforcing shall extend for approximately 1 1/2 inch from the outside corner over each surface. The case shall be designed to provide ease in installation and removal of the printing equipment.

*3.9.1 Lid and case. The case lid and panels shall be securely fastened with captive screws to prevent loss. Rubber tape, or equally suitable material, shall be permanently provided at the separation point between the case and the lid to provide a moisture proof seal when assembled.

*3.9.2 Handles. Six flush handles shall be located on the printer case, 2 on each of 2 opposite sides, and 1 on each of the other sides. The handles shall be located for placing the load in a natural, well-balanced, carrying position.

*3.10 Electrical characteristics. The printer shall be capable of operation on 105-125 volt, 60 Hertz single-phase alternating current. Selection of electrical components shall be made which will permit safe operation on 50 Hertz (Hz) current.

3.10.1 Power consumption. The power consumption of the printer shall not exceed 1200 watts.

3.10.2 Wiring. The arrangement of the electrical wiring shall be in accordance with highgrade commercial practice. The size of the wire, insulation, switch mounting, and all other electrical items and connections shall be adequate to meet the requirements specified herein.

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*3.10.2.1 Wiring diagram. When specified by the procuring activity, a permanent and legible wiring diagram on metal, metal foil, or fungus inert plastic shall be securely fastened to the printer. The lines of the diagram shall be numbered or coded to correspond with the numbering or coding of the wires.

*3.10.2.2 Symbols and electrical references. All electrical symbols and all electrical reference designations shall conform to USA Y32.16

3.10.2.3 Wiring identification. The identification of circuits shall be in accordance with the color coding of MIL-STD-681.

3.10.3 Electron tubes. The choice and application of electron tubes and shields for electron tubes shall be in accordance with MIL-STD-200.

3.10.4 Insulation. All electrical wiring shall be suitably insulated or shall be enclosed in a conduit of a material resistant to any photographic chemical solution with which it may come in contact during normal operation.

*3.10.5 Power cable. A power cable, approximately 10 feet long, conforming to MIL-C-5756, shall be provided. The wall end shall be provided with a plug conforming to MIL-C-3767/12 or MIL-C-3767/13 and the machine end shall be provided with a cord end receptacle Hubbell 5269 or equal. The wire size shall be sufficient to handle the load of the printer. The third wire shall be for grounding the equipment. The connection to the printer shall be through a recessed receptacle Hubbell 5278 or equal.

3.10.6 Controls and circuits. All electrical wiring, manner of connecting assembly and selection of components shall be in accordance with established highgrade electrical manufacturing practices. Whenever practicable, components shall be plug-in, hermetically sealed types.

3.10.7 Constant voltage transformer. A constant voltage transformer of adequate capacity with 95 to 125 volt primary, and 115 volt secondary, shall be used to power the printer lamphouse. A power switch shall be located on the control panel for cutting out the constant voltage transformer when the printer is operated on 50 Hz current.

3.10.8 Motors. Motors operating from alternating current shall comply with the requirements of CC-M-636 and NEMA Standard MG-1.

*3.11 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 govern changes in the manufacturer's part number.

*3.12 Dimensions. The over-all dimensions of the printer, ready for operation, and the dimensions of mounting, brackets, and other features such as doors, clearances, etc, shall conform to Drawing 55K17647 or to the procurement model, as applicable.

*3.13 Weight. The weight of the printer shall be in accordance with Drawing 55K17647 or the procurement model, as applicable.

*3.14 Protective finish. Unless otherwise specified by the procuring activity, the printer shall be given a final protective finish for Type I surfaces in accordance with MIL-F-14072 or MIL-STD-808, with the exceptions listed in 3.14.1, 3.14.2 and 3.14.3 below. (Note: All surfaces of the printer are considered to be Type I surfaces).

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*3.14.1 "Beauty-type" hardware and trim. "Beauty-type" corrosion-resistant steel parts of this nature need only to be buffed to a smooth finish and cleaned.

*3.14.2 Exterior paint. The exterior surfaces of the printer shall have a final finish of baked, synthetic, gray, hammer-tone enamel, color 16376 per FED-STD-595.

*3.14.3 Interior paint. The interior surfaces shall have a final finish of heat-resisting black enamel, color 37038 per FED-STD-595.

3.14.4 Case. All wood parts of the case, when a case is specified, shall receive two (2) coats of sealer, conforming to TT-V-109, diluted to 30 percent with organic solvents, and one or more coats of enamel, conforming to TT-E-529, class A, prior to assembly. The color of the case shall be as specified by the procuring activity.

3.15 Operational markings. A threading diagram showing the entire threading path of the positive and negative materials shall be installed on an outside surface of each printer. It shall be readily available for reference purposes.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The printer, including all parts and accessories, shall be fabricated and finished in a workmanlike manner with particular attention to burrs, sharp edges, accuracy of dimensions, thoroughness of soldering, welding, painting, and wiring, alignment of parts, and tightness of assembly screws, bolts, etc.

3.17.1 Dimensions and tolerances. Where dimensions and tolerances are not specified, and where they may affect the interchangeability, operation or performance of the printer, they shall be held or limited accordingly.

3.17.2 Riveting. Riveting operations shall be carefully performed to insure that rivets are tight and satisfactorily headed.

3.17.3 Screw assemblies. Assembly screws, bolts, and nuts shall be tightened sufficiently and secured so that they will not become loose due to vibration incident to the environment in which the printer is operated. The manner in which they are secured shall not prevent their removal for maintenance or repair of the printer.

3.17.4 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots or other irregularities. Where required for accuracy, gear assemblies shall be free from backlash.

3.17.5 Preparatory cleaning. Prior to application of any finish process, all surfaces shall be cleaned free from dirt, grease, oil, solder-flux, visable rust, scale, or oxides, and all other debris that might interfere with the adherence of the finish.

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*4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (does not include preparation for delivery) (See 4.4)
b. Quality conformance inspection (See 4.5)

*4.3 Test conditions

4.3.1 Temperature and humidity. Unless otherwise specified, the tests shall be conducted under prevailing room ambient temperature, relative humidity and barometric pressure.

4.3.2 Rated lamp voltage. Lamps having a rated voltage of 115 volts shall be used during all tests.

4.3.3 Line voltage. Unless otherwise specified, the line voltage during the test shall be 115 $\pm$ 2 volts, 60 Hertz, single-phase, alternating current.

4.3.4 Test materials and processing.

4.3.4.1 Materials for test. The photographic paper used for all tests, except resolution test, shall be type IV, subtype E, according to MIL-P-4672. The film duplicating material to be used on the resolution and fogging tests shall be in accordance with MIL-F-32, designation 14-14-24d.

4.3.4.2 Developing. The paper shall be developed in Armed Forces Developer No 25 in accordance with MIL-D-4827 diluted 1:2 for 1½ minutes at 68°F. The film shall be developed in Armed Forces Developer No 3 in accordance with MIL-D-4826. Developer time shall be 4 minutes with the developer at 68°F. Equivalent time shall be used if the developer solutions are used at different temperatures.

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4.3.4.3 Neutral test filter. The neutral test filter shall be a Wratten 96 neutral density filter, or equal, having a density of 1.00 ± 0.05. One neutral test filter shall be placed at the exposure reading station of the automatic exposure control system, and another neutral test filter shall be placed in the filter carrier and inserted into the light beam when conducting the basic exposure tests.

4.3.4.4 Negative halftone screen. A negative halftone screen, consisting of clear dots or clear squares on a black field of film, shall be used for testing contrast. The halftone screen shall have from 100 to 133 lines per inch, shall be free from curl, scratches, and other imperfections, and shall be not less than 9½ by 48-inches in size. It shall be spliced midway into a 9½-inch roll of aerial negatives having a length equivalent to machine maximum capacity.

4.3.4.5 Resolution test negative. A 9½-inch resolution test negative shall be used for the resolving power test. The resolution test negative shall contain high contrast targets positioned in the 9- by 18-inch format as referenced in paragraph titled "Resolving power target" of MIL-STD-150. The target patterns on the test negative shall cover a range of at least 10 to 150 lines per millimeter. The dark area of the test patterns shall have a photographic density which is at least 2.0 greater than the density of the light area. The individual targets shall be made of dark lines on a light background.

4.3.4.6 Densitometer. The densitometer used to determine the transmission density of the neutral filter and the reflection density of the prints shall be the Ansco-McBeth type, or equal. The instrument shall read zero for a transmission density when a reading is taken with no film in the machine, or the reading taken under these conditions shall be subtracted from or added to the reading taken when there is film in the instrument. The instrument shall read zero when a piece of photographic paper, not developed but fixed out, is placed in the instrument and a reflection density reading taken. If the instrument does not read zero, the reading shall be added to or subtracted from those taken on the prints. These zero readings shall be taken before each group of readings on the film or the paper are taken, and used only for the readings taken immediately after the zero readings are recorded.

*4.4 Preproduction inspection. This inspection shall consist of all inspections/tests specified herein.

4.4.1 Preproduction sample. The preproduction sample shall be one printer representative of the production model printer.

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4.5 Quality conformance inspection. This inspection shall consist of the following inspections:

a. Individual examinations/tests (see 4.5.1).
b. Sampling examinations/tests (see 4.5.2).

4.5.1 Individual examinations/tests. Each printer shall be subjected to the examinations/tests as indicated in Table I.

TABLE I

INDIVIDUAL EXAMINATIONS/TESTS		
Examination	Req Par	Insp Par
Visual Examination		4.6.1
Miscellaneous Defects	3.5.4	4.6.2
Panel Lights and Rheostat	3.8.1.3 & 3.8.5.5	4.6.3
Color Temperature	3.8.2.7	4.6.4
Tracking	3.8.3	4.6.5
Exposing Override Control	3.8.5.2	4.6.6
and Emulsion Sensitivity Control	3.8.5.3	
Drive Speed and Control	3.8.5.4	4.6.7
Wattage	3.10.1	4.6.8
Voltage Regulation	3.10.7	4.6.9

4.5.2 Sampling examinations/tests. Sampling examination/test shall consist of the examinations/tests specified in Table II. Sampling for these examinations/tests shall be in accordance with MIL-STD-105 general inspection level II, utilizing the single sampling plan.

TABLE II

SAMPLING EXAMINATION/TEST		
Examination	Req Par	Insp Par
Thin Base Materials	3.5	4.6.10
Uniformity of Illumination	3.5.6	4.6.11
Contact	3.5.7	4.6.12
Resolution	3.5.8	4.6.13
Fogging	3.8.1.2, 3.8.1.3	4.6.14
Temperature Rise	3.8.2.3	4.6.15
Automatic Exposure Control	3.8.4	4.6.16

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4.5.2.1 Rejection and retest. When a printer, selected for sampling inspection, fails to meet the requirements, the entire lot represented by the sample shall be considered as failing. Acceptance of the lot shall be withheld until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction an additional lot sample shall be taken and all tests repeated. Individual examination may be continued; however, final acceptance and shipment will be withheld until the sampling reinspection has shown that the corrective action was effective.

4.6 Inspection Methods

4.6.1 Visual examination. Each printer shall be visually examined for defects in materials and workmanship. For this examination, workmanship shall include, but not be limited to, such things as ease of operation of manual controls/ adjustments, operational markings, finish, identification marking, cleanliness and completeness of assembly.

4.6.2 Test for miscellaneous defects. At least 200 feet of negative and positive material shall be run completely through the printer and examined for scratches, abrasions, and other mechanical defects, the presence of which shall be cause for rejection.

4.6.3 Panel lights and rheostat test. While the printer is operating as above, the panel lights shall be examined for proper illumination of controls and the panel light rheostat shall be tested for compliance with 3.8.5.5 by operating the control. The intensity of the panel lights should vary as the rheostat control is operated. Any malfunction shall be cause for rejection.

4.6.4 Color temperature test. While the printer is operating as above, measure the color temperature at the printing aperture with approved color temperature meter. Any deviation above or below the specified temperature (see 3.8.2.7) shall be cause for rejection.

4.6.5 Tracking test. While printer is operating as specified in 4.6.3, the positive and negative material feed, take-up, and transporting mechanism shall be examined for proper tracking of photographic material. Any evidence of improper tracking shall be cause for rejection.

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4.6.6 Exposure override and emulsion sensitivity controls. The control shall be tested by operating the double offset knob while looking through the printing aperture. The variable density wedge should rotate. Any malfunction shall be cause for rejection.

4.6.7 Drive speed and control test. The printer, fully and properly loaded, shall be tested by operating with the speed control set at 60 fpm. Hold a tachometer against film and read the speed of travel. Repeat the test for settings of 20 fpm and 5 fpm. The actual film speed shall be within ± 1% of the specified speed. Deviations greater than ± 1% shall be cause for rejection. The materials used may be fogged, unprocessed and reused.

4.6.8 Wattage. The wattage shall be measured with a wattmeter to determine compliance with 3.10.1. Failure to comply shall be cause for rejection.

4.6.9 Voltage regulation. The constant voltage transformer shall be tested for compliance with 3.10.7 by operating the printer from inputs of 95 volts, 60 Hz, single-phase and 125 volts, 60Hz, single-phase, alternating current. Operation shall not be affected at either voltage.

*4.6.10 Test for thin base materials. The printer shall be tested to determine that it satisfactorily exposes, transports, and spools thin base (0.003 inch, or less) aerial negative material. A 200-foot roll of 9½-inch thin base aerial negative shall be printed using the specified paper and developer. During the printing run, the operation of the printer shall be observed to see that the aerial negative and printing material are properly transported through the printer. The resultant prints shall be examined to determine that good contact was maintained throughout the entire 200-foot roll. Any evidence of noncompliance shall be cause for rejection.

4.6.11 Uniformity of illumination test. Uniformity of illumination shall be tested to determine that proper evenness of illumination across the printing aperture has been established. The tests shall be performed by exposing 9½-inch roll materials using the specified paper and developer. The flash exposure of the roll paper shall be that required to give an average print density of approximately 0.70 ± 0.05. The flash print shall be at least 5 feet long and shall be processed under controlled conditions to obtain uniform development over the entire strip. Evidence of uneven illumination shall be cause for rejection.

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4.6.12 Contact tests. The printer shall be tested to determine that proper contact is established between negative and positive materials by printing the specified test negative halftone screen using the specified paper and developer. The roll of film with the negative halftone screen specified in 4.3.4.4 shall be printed at the 3 fixed speeds of 5, 20, and 60 feet per minute. One print shall be made at each of the three fixed speeds at the proper exposure to give uniform screen prints. Each screen print shall be closely examined for evidence of poor contact. Any dark areas on the prints indicate areas of poor contact, and any distortion of the dot pattern of the halftone screen indicates slippage between negative and positive materials. If either of the above conditions exists, or if the dots merge or become indistinct on any of the screen prints, the printer shall be rejected.

4.6.13 Resolution test. The printer shall be tested to determine that it produces acceptable resolution. The printer will be tested with the $9\frac{1}{2}$-inch resolution test negatives while using the specified film and developer. The test print produced from the test negative shall give a resolution reading of at least 90 lines per millimeter anywhere within the 9- by 18-inch target format. All test prints shall be read under magnification selected to produce best resolution readings.

4.6.14 Fogging test. The printer shall be tested to determine that it satisfactorily protects the printing material from fogging due to any stray light emitted by the printing lamp, negative scanning lamp, and control panel lights. While operating under darkroom conditions, the specified paper shall be threaded from the supply spool to the take-up spool. A roll of $9\frac{1}{2}$-inch aerial negative shall also be threaded from the negative supply to the negative take-up. All power switches, with the exception of the film transport, shall be turned on and the printer shall be allowed to remain in this state for at least 2 minutes. At the end of the test period, the paper shall be removed and developed in the specified developer. The print shall not have a fog density of more than 0.15. The test shall be repeated, using the specified duplication film and developer, and the print shall not have a fog density of more than 0.08. A minimum of twenty-five (25) readings shall be taken at random with equal distribution over the entire area of the print to assure the above fog density limits are not exceeded. Fog densities greater than those specified shall be cause for rejection of the printer.

4.6.15 Temperature rise. A test film having a uniform density of 2.0 or greater shall be placed in the film carrier. The printer shall be operated intermittently for a period of 60 minutes, cycled 10 seconds on and 10 seconds off. At the end of this period, the temperatures on the face of the negative shall be determined by the use of a suitable thermocouple of 0.00396 or 0.00353 inch diameter wire (No 38 or 39 AWG), which should be held or cemented to the emulsion side of the test film at the center of the aperture. The temperature in degrees F shall be determined with an instrument accurate to $\pm$ 4°F. Temperature shall not exceed 150°.

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4.6.16 Automatic exposure control test. The automatic exposure control shall be tested by inserting different sets of neutral density test filters into the negative reading station and filter carrier, and producing flash exposure prints using the specified paper and developer. The automatic exposure control shall be adjusted so that a neutral density filter of 1.0 will give a print density of 0.70 ± 0.05 with the exposure wedges set at approximately midrange. At this setting the automatic exposure control shall be used to produce prints from neutral densities of 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, and 2.1. All flash prints produced from this series of neutral densities shall be 0.70 ± 0.10. Failure of any one of the flash prints to have the required density shall cause rejection of the printer.

4.6.17 Environmental testing. The following environmental tests shall be conducted as preproduction tests only.

4.6.17.1 Elevation test. The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

a. With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.6.13	Resolution
4.6.14	Fogging

b. Reduce the barometric pressure to 20.6 +0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.6.17.1a. The equipment shall meet specified performance for those measurements.

c. Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

d. Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.6.17 1a. The equipment shall meet specified performance for these measurements.

4.6.17.2 Moisture-resistance test. The equipment shall be tested as follows:

4.6.17.2.1 Test conditions.

a. Do not move equipment from the humidity chamber for measurements.

b. Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

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4.6.17.2.2 Test Method

a. Dry at 130° ± 5°F for 24 hours.
b. Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.
c. Take measurements as specified in 4.6.17.2.3.
d. Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.6.17.2.3 shall be made during the periods shown on the standard and shall comply with 4.6.17.2.3.
e. After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.6.17.2.3.

4.6.17.2.3 Performance. Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.6.13	Resolution	Full specification requirements
4.6.14	Fogging	Full specification requirements

4.6.17.2.4 Failure. If any equipment fails to meet the performance specified in 4.6.17.2.3 during cycling, it does not pass the test.

4.6.17.3 Temperature. The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169 except the temperature at step 4 shall be ± 115°F and at step 9, +40°F. Performance shall be as follows:

Step	Paragraph	Measurement	Performance
1, 4, 5, 9, 10	4.6.13	Resolution	Full specification requirements
	4.6.14	Fogging	

4.6.17.4 Leakage current. Following the extreme temperature tests, and immediately after the equipment has returned to prevailing room temperature and pressure all conductors shall be tested for leakage current. Leakage current shall be determined by the application of an alternating-current power source having a potential of 115 volts ± 1 volt, 60 Hertz, being applied between the "hot" positive line conductors and ground. All adjacent metal parts not in the circuit under test shall be considered as ground. Any electrical component load across the test voltage shall be disconnected by opening the circuit at the grounded conductor. Radio noise filters, lamps, and plug-in thermal relays may be removed for this test. A test of all circuits, taken at the power input connector, and resulting in a reading of 2 milliamperes or more shall be cause for rejection.

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4.6.17.5 Test for internal vibration. Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.7.1d:

a. Secure the equipment (shock mounts, if any, removed or blocked) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall not be less than 0.030 inch.

c. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

e. Upon completion of the vibration cycles, the printer shall be operated by running 200 feet of material through it. There shall be no impairment of operation nor malfunction of components as a result of the vibration cycles.

4.6.17.6 Dielectric. All conductors shall be tested for dielectric strength. A test voltage shall be applied between all conductors and ground. All adjacent metal parts not in the circuit under test shall be considered as ground. The test voltage shall have a frequency not greater than 100 Hertz and a root-mean-square-value (RMS) of twice the rated peak operating voltage plus 1000 volts. Where maximum peak operating voltage is greater than 700 volts, the root-mean-square test voltage shall be 1000 volts greater than 1.5 times the maximum peak operating voltage. The applied test voltage may be alternating or direct current. Any electrical component load across the test voltage shall be disconnected by opening the circuit at the grounded conductor. Radio noise filters, lamps and plug-in thermal relays may be removed for this test. The test voltage shall be applied for 1 minute to all conductors without damage, insulation breakdown, deterioration, or flashover.

4.6.17.7 Suppression. The printer shall be tested in accordance with MIL-STD-462 for compliance with 3.7.2.

4.6.17.8 Fungus. The equipment and assemblies shall be subjected to fungus test specified in Method 508 of MIL-STD-810.

4.6.18 Inspection of preparation for delivery. Preservation, packaging and packing shall be inspected to insure compliance with the requirements of the applicable specifications referenced in section 5.

*5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

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5.1.1 Level A. The printer shall be preserved in accordance with Method II of MIL-P-116. Hardware shall be packaged in cloth bags conforming to PPP-B-35 or aluminium foil bags conforming to MIL-B-131. Contact preservative is not required on cadmium plated items. The bagged hardware shall be packaged and stored in separate packs so the unit quantities will be sufficient in number to assemble one complete printer.

5.1.2 Level C. The printer shall be preserved and packaged in accordance with manufacturers commercial practice.

*5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each printer, as specified in 5.1.1, shall be packed in exterior shipping containers conforming to MIL-B-26195, Class 2 for overseas shipment. Exterior containers shall be of uniform shape and size, be of minimum cube and tare consistant with the protection required. The gross weight of each pack shall be limited to Table III of MIL-B-26195. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B. Each printer, as specified in 5.1.1, shall be packed in exterior shipping containers conforming to MIL-B-26195, Class I for domestic shipment. Exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to Table III of MIL-B-26195.

5.2.3 Level C. Each printer, as specified in 5.1.2, shall be packed for shipment in commercial exterior containers in such a manner that will insure acceptance by common carrier and safe delivery at destination. Containers and closures shall comply with the uniform Freight Rule or Regulations of other carriers as applicable to the mode of transportation.

*5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The EN-6B continuous printer is intended for continuous printing 70mm, 5- and 9½-inch wide rolls of aerial film on roll photographic printing papers and film duplicating materials. The printer is to be operated in an automatic manner and intended to be used where large quantities of high quality, first phase, photographic prints, and duplicate negatives are required from rolls of aerial film on the basis of equal numbers of prints from each and every negative on the roll.

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6.1.1 Normal service-life. The anticipated normal service-life of the printer is expected to be 1 year while operating 12 hours per day. This information is provided as guidance to the contractor in determining the quality and durability of the components.

6.1.1.1 Conditions of field use. Conditions of field use may be defined as follows:

a. Air temperature range of 44°F to 95°F.
b. Relative humidity 5 to 100 percent.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Maximum capacity of roll material (see 3.5).
c. Type of filter system required (see 3.5.3.1).
d. Whether production model is to be made available (see 3.6).
e. Whether fungus proof tests are required (see 3.4.2).
f. Whether wiring diagram (attached to equipment) is required (see 3.10.2.1).
g. Whether special marking is required (see 5.3).
h. Whether case is required and color of case (see 3.9 and 3.14.4).
i. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

*6.3 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - AS
Army - EL
Air Force - 70

Review activities:
Navy - AS
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Air Force - 70

Preparing activity:
Air Force - 70

Project No: 6740-0143

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Hill Air Force Base, Utah 84401

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R25$

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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